

Sana Saab

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EDUCATION

Brown University, *Sc.B. Computer Science, 4.0/4.0 GPA*

Providence, RI | **Expected Graduation May 2027**

Relevant Courses: Foundations of AI, Introduction to Software Engineering, Statistical Inference, Program Design with Data Structures and Algorithms, Systems Fundamentals, Interaction Design, Database Management Systems, Algorithms

SKILLS & INTERESTS

Programming Languages: Python, Java, Racket, ReasonML, TypeScript, JavaScript, CSS, C, C++, Assembly, SQL

Technologies and Frameworks: Linux, Firebase, React Native, React, SQL, Git, Node.js, Spark, Flyte, GraphQL, gRPC, AI/ML

TECHNICAL EXPERIENCE

LinkedIn, *Software Engineering Intern*

Mountain View, CA | May 2026 – August 2026

- Designed and implemented a writer-editor QA loop for a generative-AI pipeline that transforms job market anomalies and insights derived from 40M+ member profiles and 7M+ job postings into cited articles, increasing publish-ready outputs (scoring 18+/20 on a quality rubric) from 5% to 90% in a 56-article comparison
- Independently shipped the project end-to-end, developing a gRPC-to-GraphQL backend bridge and Remix frontend and resolving routing, access-control, and deployment issues across 3 pipelines and 4 services to production
- Designed JUDE/MUSE recommendation experiments and built a Spark/Flyte Semantic ID pipeline processing 677M+ unique members to evaluate tradeoffs between candidate-job relevance and predicted engagement

Amazon Web Services (AWS), *Software Engineering Intern*

Boston, MA | May 2025 – August 2025

- Developed a native Oracle redo log parser connector for Debezium CDC as an alternative to LogMiner
- Built core features including transaction grouping, before/after column image reconstruction, system noise filtering, schema-aware type interpretation, transaction ID parsing, and SCN parsing to support AWS's Data Migration Services
- Delivered a system with performance gains over LogMiner by reducing SQL overhead and session management errors

Brown Daily Herald Mobile App, *Backend Developer*

Providence, RI | January 2024 – January 2025

- Collaborated with a team of 20 software engineers and designers to develop [a mobile application](#) supporting 1000+ active users using React Native, Cloud Functions for Firebase, and Expo Push Notifications to deliver real-time news updates
- Implemented, tested, and deployed 7+ Firebase Functions in TypeScript to handle backend operations such as creating users and updating user settings. Integrated security features like API throttling to safely support a growing userbase

Introduction to Software Engineering, *Head Teaching Assistant*

Providence, RI | April 2025 – Present

- Lead staff of 10 TAs, conducting hiring, interviews, onboarding, training, and weekly coordination meetings
- Collaborated with professors to redesign the course structure and materials to reflect real-world software engineering by creating new content on AI-assisted coding, using Zod with refinements, and integrating with large, pre-existing codebases
- Supported 100+ students through coding assignments, labs, and projects via in-person office hours and an online board

TECHNICAL PROJECTS

Distributed Key-Value Store, *C++*

Providence, RI | Spring 2025

- Built a horizontally-shardable, fault-tolerant key-value store in C++, implementing a custom socket-based RPC layer and a shard controller for dynamic join, leave, and move operations across servers, exposing client-facing operations
- Replaced a global lock with a fine-grained concurrency model (60-way bucket-sharded hashmap, per-bucket reader-writer locks), boosting throughput ~5x over the baseline (2,116 to 10,814 ops/sec on puts; 2,037 to 10,938 ops/sec on gets)

Go Agent, *Python*

Providence, RI | Fall 2024

- Developed advanced AI agents for the board game Go using Monte Carlo Tree Search (MCTS), Minimax, Alpha-Beta pruning, and neural network-based techniques, and compared their performances to choose an optimal base agent
- Implemented a transposition table system to cache and reuse previously evaluated game states, improving search depth
- Optimized an advanced agent by integrating iterative deepening search (IDS) with transposition tables, achieving a consistent +10.0 score improvement over baseline IDS and maintaining a 100% win-rate against it

COMMUNITY INVOLVEMENT

Brown Dabke Club, *Founder, President*

Providence, RI | September 2024 – Present

- Founded Brown Dabke Club, tripling practice attendance and growing membership to 50+ students within a few weeks
- Recruited and managed a 7-person executive board, overseeing weekly meetings, finances, and member duties
- Organized and led over 5 workshops, including a 500+ attendee event, boosting sign-ups and community engagement

St. Jude Children's Research Hospital, *Volunteer Coordinator*

Providence, RI | January 2019 – August 2023

- Completed over 100 hours of service organizing and working events; helped raise over \$1M for St. Jude Children's Hospital